F1 RACING ANALYTICS

PROJECT REPORT

Formula 1 Data Visualization with Tableau and Flask

Turning racing data into insights

AninteractiveFormula1 analytics project that bringsdriverperformance, constructor results, tyre statistics, circuit information and lap-time analysis together in a web-accessible Tableau dashboard.

Project	F1 Racing Analytics
Visualization	Tableau Public
Web application	Python Flask
Deployment	Vercel
Source code	GitHub - sudhanshumedicaps

Live website: https://f1-project-mauve.vercel.app/

Tableau dashboard: https://public.tableau.com/views/F1_racing_17936297010240/F1RACINGANALYTICS?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link Prepared for academic project presentation

1. Project Overview

F1RacingAnalytics is a datavisualization project built to present Formula 1 racing statistics through an interactive Tableau dashboard. The dashboard brings several views of racing performance into one place, helping users explore drivers, constructors, race wins, championships, tyre analysis, circuits and lap-time patterns.

The Tableau dashboard is published on Tableau Public and embedded in a website created with Python Flask. The website providesasimpleway to accessthe dashboard through a browser.

2. Project Aim and Objectives

ThemainaimistoturnFormula1dataintovisualinformation that is easier to explore and understand.

Objectives

- Present key summary indicators such as total circuits, total drivers, total wins and average lap time.
- Compare driver performance through wins, rankings and championship statistics.
- Explore constructor standings and team performance.
- Visualize tyre-related statistics, circuit information and lap-time analysis.
- PublishtheTableaudashboard through a web application so it can be accessed online.

3. Project Features

Feature	Purpose
KPI cards	Summarize headline metrics: circuits, drivers, wins and average lap time.
Driver analysis	Explore driver results, top drivers, wins and championship counts.
Constructor standings	Compare the performance of Formula 1 constructors.
Tyre analysis	Present tyre-related comparisons and patterns.
Circuit map and analysis	Explore circuit locations and circuit-level statistics.
Lap-time analysis	Visualize lap-time information and comparisons.

4. Dashboard Overview

TheTableaudashboard is themainanalyticalcomponent of the project. It combines summary KPI cards and multiple charts in a single view. The dashboard includes the following sections:

- F1 Drivers
- Top 10 Drivers
- Constructor Standings
- Wins by Drivers
- Championships by Drivers
- Tyre Analysis
- Circuit Map
- Lap Time Analysis
- Circuit Analysis

Dashboard KPIs

KPI	Displayed value
Total Circuits	29
Total Drivers	19
Total Wins	313
Average Lap Time	80.25

The KPI values above are taken from the project dashboard. Average lap time is reported as displayed in the dashboard; its unit or aggregation definitionwas notprovidedin theproject details sharedfor this report.

5. Tools and Technologies

Technology	Use in the project
Tableau	Creating charts, KPI cards and the interactive dashboard.
Python	Application programming language.
Flask	Serving the web page that embeds the Tableau dashboard.
HTML / CSS	Structuring and styling the website interface.
Vercel	Hosting and deployment of the web application.
Git / GitHub	Version control and source-code repository.

6. Application Architecture

The projecthastwomainparts: theTableauvisualization layer and the Flask web layer.

Data and visualization: Formula 1 data is explored in Tableau, where worksheets and dashboard elements present the analysis.

Web application: A Flask application serves an HTML page. The page embeds the published Tableau Public dashboard, allowing visitors to view the visualization through the project website.

Deployment:The project sourcecodeis maintained in GitHub and the web application is deployed on Vercel.

7. Key Dashboard Insights

Thedashboard isdesignedto support explorationof several aspects of Formula 1 performance. The available project details identify the following analysis areas:

- Driver performance can be explored through rankings, wins and championship counts.
- Constructor standings provide a team-level comparison.
- Tyre analysis presents patterns related to tyre use.
- The circuit map and circuit analysis provide a geographical and circuit-level view.
- Lap-time analysis allows comparison of lap-time information.

These are the analytical questions the dashboard enables users to explore. Specific findings or statistical conclusions have notbeen addedwhere supporting chart valueswerenot provided.

8. Website and Project Links

The FlaskwebsiteembedstheTableauPublicdashboard and acts as the public-facing entry point to the project.

Live website: https://f1-project-mauve.vercel.app/

Tableau Public: (Mentioned Above)

GitHub repository:https://github.com/sudhanshumedicaps/f1-project

9. Limitations and Future Improvements

The exact source dataset, date coverage, fielddefinitions andcalculation logic for each visualization were not included in the project details available for this report. Therefore, this report does not state dataset row counts, season coverage or calculated findings beyond the KPI values shown on the dashboard.

Possible future improvements include documenting the source dataset and field definitions, adding explanations for each KPI and chart, providing more filter controls, and including a methodology section describing calculations and data cleaning.

10. Conclusion

F1Racing Analytics combines Tableau-based data visualization with a Flask web application to make Formula 1 statistics accessible through an online dashboard. The project demonstrates a workflow that connects data exploration, dashboard development, web integration, version control and deployment.